



FIRST SEMESTER 2026-2027

Course Handout (Part II)

Date: 10/08/2026

In addition to part I, (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No : MATH F214/ MAC F214
Course Title : Elementary Real Analysis
Instructor-in-Charge : SANGITA YADAV (sangita.yadav@pilani.bits-pilani.ac.in, 6067K)
Team of Instructors : Sangita Yadav, Gaurav Dwivedi (gaurav.dwivedi@pilani.bits-pilani.ac.in, 6067J)
Tutorial Instructors : Sangita Yadav, Gaurav Dwivedi

1. Course Description: Countable and uncountable sets, Real numbers and their properties, Sequence of real numbers, Continuous functions of real variable, Metric space, Compact set, Connected set, Completeness, Continuous function on a metric space, Riemann integration, Sequence and series of functions.

2. Scope and Objective of the Course: This course aims to discuss several properties of real numbers and functions of real variables. The notions of metric spaces will be introduced and a few concepts in analysis, such as continuity, the convergence of sequences, the convergence of sequences and series of functions, will be discussed. We will also discuss the rigorous theory of Riemann integration. Students will also learn to read and write rigorous proofs so that they can communicate mathematical ideas effectively. This course is also a basic course to build background for other advanced courses in mathematics.

3. Course Learning Outcome:

CLO1. Students will learn about real numbers and their elementary properties, sequences of real numbers, and the generalization of distance to abstract sets.

CLO2. Students will be introduced to the notion of convergence of series and sequence of functions.

CLO3. Students will study the rigorous theory of integration.

4. Textbooks:

[TB1]. Walter Rudin, Principles of Mathematical Analysis, McGraw Hill Education (India), 3rd Edition, 2023.

[TB2]. Kenneth A. Ross, Elementary Analysis: The Theory of Calculus, Undergraduate text in Mathematics, Springer, 2nd Edition, 2013.

5. Reference Books:

[RB1]. Robert G. Bartle, Donald R. Sherbert, Introduction to Real Analysis, Wiley India Edition, 4th Edition, 2011.

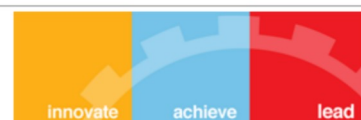
[RB2]. Tom M. Apostol, Mathematical Analysis, Narosa, 2nd Edition, 2002.

[RB3]. Terrence Tao, Analysis I and II, Hindustan Book Agency, 2017.

[RB4]. Paul Zorn, Understanding Analysis, CRC Press, 2nd Edition, 2018.

6. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1. Preliminaries and Logic	L1-L2: Revision of functions, sets, and logic, some methods of proofs	1.1-1.5 [RB4]	A quick revision of functions, set theory, and logic





2. Real numbers	L3-L6: Construction of real numbers, the set of real numbers as an ordered field, extended real numbers, Infimum and Supremum, Archimedean property	1-6 [TB2]	Students will learn about real numbers and some of their elementary properties.
3. Cardinality of a set	L7-L8: Countable and uncountable sets	2.1-2.14 [TB1]	Students will learn about the cardinality of an infinite set.
4. Sequences and series	L9-L17: Sequences, Limits of a sequence, monotone and Cauchy sequence, subsequence, limsup and liminf, Series, the decimal expansion of real numbers	7-12 [TB2], 14-16 [TB2]	A thorough study of sequences of real numbers and their properties.
5. Metric Spaces	L18-L22: Metric Spaces, Compact sets, Perfect sets, connected sets, Cantor set L23-L25: Sequence in metric space, Subsequence, Cauchy sequence, complete metric space L26-L30: Continuous functions, Uniform Continuity, continuity and compactness, continuity and connectedness.	2.15-2.47 [TB1] 3.1-3.14 [TB1] 4.1-4.24 [TB1], 17-20 [TB2]	A generalization of concepts of distance to abstract sets.
6. Sequence and Series of functions	L31-L33: Point-wise and uniform convergence of sequence and series of functions and related properties of integrability	7.1-7.18 [TB1]	Students will be introduced to the notion of convergence of series and sequence of functions.
7. Integration	L34-L40: The Riemann Integral and its properties	32-35 [TB2]	Students will study the rigorous theory of integration.

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Continuous Evaluation*		21%	TBA	CB
Mid-Semester Test	90 Min.	34%	As announced by AUGSD	CB
Comprehensive Examination	180 Min.	45%	As announced by AUGSD	OB/CB

*4 announced quizzes each of 7% weightage. The score of the best three quizzes will be considered for grading.

8. Criterion for NC: The student with total marks below 7.5% of the course total will be awarded NC grade.

9. Chamber Consultation Hour: Saturday 11:00 AM--12:00 PM.

10. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

11. Notices: All the notices related to the course will be posted on NALANDA.

12. Make-up Policy: Make-up for mid-semester test and comprehensive exam will be granted only in genuine cases. There is no makeup for the quizzes under any circumstances.

Sangita Yadav
Instructor-in-charge
MATH F214/MAC F214

